

Golden section: shrinks by φ^{n-1} after n queries

 l_1

 $l_2 = l_1 \varphi^{-1}$

 $l_3 = l_1 \varphi^{-2}$

 $l_4 = l_1 \varphi^{-3}$

 $l_5 = l_1 \varphi^{-4}$